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The Thermal Decomposition of Ammonium Perchlorate. I. Introduction, Experimental, Analysis of Gaseous Products, and Thermal Decomposition Experiments

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Proc. R. Soc. Lond. A 1954 **227**, 115-132
doi: 10.1098/rspa.1954.0284

The thermal decomposition of ammonium perchlorate

I. Introduction, experimental, analysis of gaseous products, and thermal decomposition experiments

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(*Communicated by H. W. Melville, F.R.S.—Received* 21 *July* 1954)

The thermal decomposition of ammonium perchlorate *in vacuo* and under small initial pressures of nitrogen, to suppress sublimation, has been investigated in the temperature ranges 220 to 280° C and 380 to 450° C. The experimental techniques for following the decomposition and subsequent analysis of the products are described and gas analysis results given.

In the low-temperature range only 30 % decomposition occurred though the 'residue' was still ammonium perchlorate. *In vacuo* sublimation occurred all the time and also after decomposition had ceased which indicated that the reaction was not in the vapour phase. Some of the properties of the sublimed material and the 'residue' were investigated; in particular, it was found that the residue which was porous in texture (the decomposition had occurred throughout the crystal) could be 'rejuvenated' by exposure to a solvent vapour.

The crystal transformation at 240° C from orthorhombic to cubic, the addition of impurities which might be intermediate decomposition products, and the addition of some metallic oxide catalysts, were also investigated.

1. Introduction

The thermal decomposition of ammonium perchlorate is 'unique' in character for though it is obviously of the general type of solid decomposition,

$$\text{solid } A = \text{gas } B + \text{gas } C + \text{gas } D + \ldots.$$

it is also, under most of the present investigation conditions, of the second type,

$$\text{solid } A = \text{solid } B + \text{gas } \ldots.$$

This is due to the fact that under high vacuum or in a gas stream, the decomposition ceases after approximately 30 % loss in weight has occurred. The possible reasons for this stoppage will be discussed later. This stoppage of decomposition is not abrupt, and the resulting sigmoid decomposition curves are similar to those obtained from reactions which go to completion. The 30 % decomposition takes place throughout the crystal and not only in a surface layer or in separate zones. The amorphous material remaining, which will be referred to as the 'residue', is pseudomorphous with the unheated salt and is still entirely ammonium perchlorate. It can thus be seen how this decomposition comes under both of the above headings, for the residue is analogous to solid B in the second type.

Most of the investigations on the thermal decomposition of solids have been on those giving at least one non-volatile solid product. Decomposition reactions of the other type are small in number. Those investigated are: ammonium perchlorate, chlorate, nitrate (where melting accompanies decomposition), nitrogen iodide, alkaline azides, mercuric oxide and mercury fulminate. In none of these decompositions does the reaction spread completely through the solid and yet leave a fraction undecomposed. The retarding effect of ammonia on the decomposition of ammonium perchlorate, ammonium nitrate and nitrogen iodide is the only similarity, and in all three the reaction proceeds again after removal of the ammonia.

Very little research has been carried out on the thermal decomposition of inorganic perchlorates with metallic cations and apparently none on their kinetics. Recently Martin & Woolaver (1945) have shown that only certain perchlorates such as those of potassium, sodium and lithium, form the chloride and oxygen on heating, while others, such as magnesium, calcium, ferric and aluminium perchlorates, form the oxide or hydroxide, perchloric acid and water.

The decomposition of ammonium perchlorate has previously been investigated by Naoum & Aufschlager (1924), who stated that its explosion products were Cl_2, N_2, O_2, H_2O and HCl, and that oxides of chlorine were also evolved during the slow decomposition at red heat. Dodé (1935, 1938) has re-investigated the nature of the products of the thermal decomposition; the work was mainly of a qualitative nature: analysis of the products over a wide range of temperature and a discussion on the influence of temperature on the oxidation products of ammonia.

Dodé reported that ammonium perchlorate was perfectly stable at 110° C, though at 130°, *in vacuo*, it started to sublime and underwent a fast decomposition, and the loss in weight observed from this moment represented the sum of the two phenomena. In the present investigation both decomposition and sublimation were negligible under 200° C. By plotting a graph of weight lost due to decomposition in 1 h over a wide range of temperature against temperature, Dodé obtained a curve presenting an anomaly at about 230° C which was attributed to a crystal transformation occurring at 240° C. In the present work this was confirmed and found to occur at 240° C.

As the temperature was raised Dodé found that the amount of decomposition predominated more and more over the sublimation until at 400° C the latter had completely disappeared. However, in the present work, under vacuum conditions, the rate of sublimation increased with temperature at a faster rate than did the decomposition rate. Also Dodé reported that small amounts of the salt (0·2 to 0·3 g) deflagrated at 380 or 390° C, while in the present work the decomposition was still sufficiently slow to be measured at 450° C without deflagration occurring.

The present results agree with those of Dodé that below 300° C the equation

$$4NH_4ClO_4 \rightarrow 2Cl_2 + 3O_2 + 8H_2O + 2N_2O$$

represented the major part of the products. Above 300° C the proportion of nitric oxide became appreciable and increasing, and at temperatures above 350° C the following equation was considered more representative:

$$2NH_4ClO_4 \rightarrow 4H_2O + Cl_2 + O_2 + NO,$$

together with the many possible addition products.

The stoppage of the decomposition reaction under vacuum conditions, after only 30 % loss in weight due to decomposition, has not previously been reported. The effect of an important, though usually neglected, variable, particle size of salt, had been investigated.

The experimental work can be divided conveniently into two parts: (i) the apparatus used to follow the decomposition, the analysis of the gaseous products and the analysis results, and (ii) the effect of various factors on the decomposition.

The reporting of the work has also been divided in this way so that the effect of various factors such as temperature, particle size, the interpretation of the pressure-time decomposition curves by many of the current theories on thermal decomposition of solids, and the discussion of the results, are to be found in a following paper (part II).

2. Experimental

(i) *Preparation and analysis of ammonium perchlorate*

The ammonium perchlorate used for this work, unless otherwise stated, was commercial quality twice recrystallized from water. Further recrystallization was found to be unnecessary, as no difference in this quality, hence known as salt *A*, or that after seven recrystallizations was found either analytically or in the respective decomposition curves. A small quantity of the salt was also prepared from A.R. 60 % perchloric acid and an excess of pure ammonia. This material was analytically the same as salt *A*. The salt was analyzed by the Kjeldahl method which gave the theoretical result for ammonia content, and also analyzed for perchlorate content by a method due to Smeets (1934), which gave results which were rather low. This was because the extraction of the sodium perchlorate formed during the analysis was always incomplete.

The residue and sublimed salt were analyzed by the Kjeldahl method and X-ray powder spectra also obtained showing both to be identical with salt *A*.

(ii) *Experimental technique used to follow the decomposition*

The decomposition was studied under three different sets of conditions necessitating separate pieces of apparatus. These conditions were (*a*) in a slow stream of nitrogen at atmospheric pressure and where the rate was measured by passing the gas stream through potassium iodide absorbers and titrating with thiosulphate the iodine liberated over various times, (*b*) in a vacuum apparatus where the decomposition in the temperature range 200 to 300° C was followed by measuring the increase in pressure of permanent gases, the others being condensed in liquid oxygen, and (*c*) at higher temperatures of 400° C and over where a small initial pressure of nitrogen was required to suppress sublimation and the decomposition was again followed by means of increase in permanent gas pressure.

The greatest part of the investigation was carried out under the second condition, as the first did not enable the operator to obtain a comprehensive decomposition-time curve. Both accumulatory (sigmoid) and rate-time curves were obtained in this apparatus. A horizontal-type tube furnace was used to heat the sample of either 100 to 150 mg of salt *A*. The temperature was measured with a chromel-alumel thermocouple and controlled to within $\pm \frac{1}{2}$° C. The sample to be decomposed was weighed into a glass tube 19 cm long × 4 mm internal diameter (sealed at one end), and kept in place at the closed end by means of a small plug of glass wool. The tube passing through the furnace was of such a length as to allow the salt in its small tube to lie out of the furnace and also to have a glass 'pusher' containing an iron core which could be moved from outside the apparatus with a magnet. The sample was thus pushed into the centre of the furnace when the required temperature and degree of vacuum had been reached and the pusher quickly withdrawn. The length of the tube containing the salt was such that with the latter in the centre of the furnace the open end protruded 3 cm out of the furnace and thus collected any sublimate. At the end of the run the weight lost due to decomposition was simply the weight loss of the tube. If desired the amount of sublimation was found by cutting the tube between sublimate and residue, weighing that containing the sublimate, washing out and re-weighing.

The apparatus was evacuated by the usual rotary oil/mercury diffusion pump system through liquid oxygen traps. Pressures were measured on a 0 to 2 mm McLeod gauge, and the dead space volume could be altered by inclusion of one or more of three calibrated bulbs. A volume of 1370 ml. was used for accumulatory pressure runs with 100 mg salt and 1910 ml. when 150 mg salt was used. The vacuum line also contained three traps which were used to collect the condensible gases and fractionate them prior to analysis.

The rate of decomposition was obtained from this apparatus by cutting off the pumps at fixed intervals and allowing the pressure to rise for 30 or 60 s when the resultant pressure was measured. The volume used in this case was 690 ml. This method of rate measurement is often criticized, as the readings must give values which are too high because of the 'running' pressure in the apparatus, and also the rate itself must vary over the period of the reading. The rate curves were not, in any case, interpreted mathematically, but only for comparative purposes. It was found that the 'running' pressure was 10 % of the rate at any time (for 30 s 'build-ups'). Using pumps of widely different pumping speeds no change was found in the running pressure, so that it was the actual pressure due to the decomposing salt and not due to the inability of the pumps to remove the accumulated gas after previous reading.

For the high-temperature decomposition a small initial pressure of nitrogen or dry air (1 to 4 cm) was let into the apparatus to stop sublimation. A short boat was used in this case and the volume of the apparatus kept small, so that the decomposition could be followed by the increase in permanent gas pressure measured by a glass spiral gauge used as a null-point instrument in conjunction with a mercury manometer.

(iii) *Gas analysis*

All the following gases were reported to be products of the thermal decomposition from previous work by Dodé: Cl_2, N_2O, N_2O_4, O_2, N_2, H_2O, HCl, ClO_2 (below 300° C) and NOCl (higher temperatures). Their presence was confirmed and, in addition, perchloric acid was found. Dodé used a very complex analysis method which apparently never yielded any complete analysis figures. In the present scheme, however, the gases were never completely analyzed for any particular run as the method used to estimate some always eliminated others. Complete analyses were obtained by using the different methods on the gaseous products from a number of runs carried out under the same conditions. The gases, in all cases collected during the vacuum runs, were analyzed in the following fractions: (*a*) oxygen and nitrogen, (*b*) chlorine and chlorine dioxide, (*c*) nitrous oxide, chlorine and nitrosyl chloride, (*d*) perchloric and hydrochloric acids.

Oxygen and nitrogen

100 or 150 mg of salt *A* was heated in the vacuum apparatus already described to which was connected a Toepler pump. All condensible gases were trapped at −180° C and the oxygen and nitrogen collected with the Toepler pump. They were then transferred to an Ambler gas analysis apparatus, exploded with pure hydrogen and the oxygen and nitrogen (by difference) determined. The results of this analysis are shown in table 1.

TABLE 1

run no.	temperature (° C)	duration (min)	total vol. N_2+O_2 at n.t.p./10 mg salt decomp.	volume percentages O_2	N_2	average O_2	average N_2
T10A	230	270	1·24	81	19		
11	230	270	1·24	79·2	20·8	80	20
12	230	270	1·23	80	20		
T 7A	240	270	1·27	86·6	13·4		
8	240	270	1·32	90	10		
9	240	270	1·26	89·5	10·5	88·4	11·6
13	240	270	1·26	86·3	13·7		
15	240	270	1·25	89·5	10·5		
V21A	250	67	—	94	6		
T 4A	250	275	1·29	94·6	5·4	93·2	6·8
5	250	275	1·33	91·9	8·1		
6	250	270	1·25	92·4	7·6		
T17A	260	270	1·31	89·7	10·3		
20	260	270	—	89·3	10·7	91	9
24	260	150	—	92·7	7·3		
31	260	115	1·27	92	8		
T30A	270	80	1·24	91·3	8·7	90·7	9·3
32	270	75	1·24	90·1	9·9		
T25A	280	60	1·26	92·7	7·3		
T29A	290	85	1·26	91·3	8·7	91·3	8·7
28	290	70	1·26	91·3	8·7		
T26A	300	60	1·23	90·4	9·6		
T33A	310	60	1·27	90·4	9·6		
T37A	322·5	30*cp**	1·22	90	10	89·7	10·3
36	322·5	60*ac*	1·23	89·5	10·5		
T35A	335	20*cp*	1·32	87·8	12·2	88	12
34	336	60*ac*	1·26	88·3	11·7		

* *cp* means a faster pumping rate than usual; *ac* means that the permanent gases were not pumped off until the end of the run.

Chlorine and chlorine dioxide

The vapour pressures of chlorine, chlorine dioxide and nitrous oxide were sufficiently high at −78° C to allow of their separation from the remainder of the products by fractionation of the gases originally condensed during the decomposition. After fractionation from -78 to $-180°$ C the gases in the lower temperature trap were treated differently according as to which gases were required. When chlorine and chlorine dioxide were required the tube containing the condensed gas, still immersed in liquid oxygen, was removed from the apparatus and fitted to another apparatus through which a stream of nitrogen could sweep the condensed gases, when allowed to warm up, through a set of absorber bulbs containing neutral potassium iodide solution. The nitrous oxide had no effect on

Table 2

run no.	temperature (° C)	duration (min)	% weight lost	% sublimed	g mol. Cl_2	ClO_2	total Cl_2	total H^+
85	224	270	26·4	6·6	0·44	0·037	0·46	—
86	224	270	26·9	7·7	0·43	0·052	0·46	—
88	224	270	26·0	6·1	0·43	0·065	0·46	—
65	226	280	27·6	4·4	0·455	0·011	0·46	—
64	228	270	28·2	7·2	0·44	0·02	0·45	—
207	230	170	32·1	—	0·44	0·02	0·45	0·13
209	230	170	32·0	—	0·42	0·038	0·44	0·173
212	230	170	33·0	—	0·434	0·022	0·44	0·14
211	230	170	32·0	—	0·431	0·023	0·44	0·16
213	230	180	32·4	—	0·44	0·02	0·45	0·153
70	234	270	29·7	9·5	0·433	0·024	0·445	—
66	235	270	29·0	7·3	0·43	0·035	0·447	—
62	236	270	29·0	10·0	0·38	0·041	0·40	—
63	238	270	25·5	6·8	0·392	0·049	0·42	—
71	241	270	27·2	7·5	0·41	0·046	0·43	—
67	244	270	26·3	9·0	0·352	0·06	0·38	—
68	250	240	26·8	16·9	0·40	0·053	0·43	—
58	251	270	28·3	11·2	0·41	0·064	0·442	—
59	253	270	28·4	16·0	0·40	0·071	0·435	—
60	256	270	27·4	24·4	0·368	0·088	0·412	—
69	260	270	26·0	22·5	0·40	0·066	0·433	—
61	277	270	28·4	37·0	0·336	0·082	0·378	—
83	281	120	23·0	38·0	0·32	0·214	0·42	—
84	281	120	21·4	46·2	0·33	0·17	0·41	—
72	285	270	27·5	42·1	0·36	0·121	0·42	—
73	305	270	19·4	80·6	0·37	0·195	0·467	—
76	327	90	14·7	85·2	0·41	—	—	—
74	331	90	14·3	85·5	0·38	0·145	0·45	—

All the above figures were taken from rate runs. The salt was type *A* except in runs 207 to 213, where it was graded particle size 3 (i). The difference between the total gram-molecule of Cl_2 and 0·5 is approximately half the total gram-ions of H^+.

this solution, but both chlorine and its dioxide liberate iodine and the dioxide also liberates a further amount on acidification of the solution. As these reactions were stoicheiometric it was possible to calculate the amounts of both these gases from the iodine liberated before and after acidification. The iodine was estimated with thiosulphate.

The reactions of chlorine dioxide and potassium iodide solutions were investigated in this connexion and the following equations found to be correct:

$$6ClO_2 + 10KI \rightarrow 4KIO_3 + 6KCl + 3I_2, \text{ in neutral solution,}$$
$$2ClO_2 + 10KI + 4H_2SO_4 \rightarrow 2KCl + 4K_2SO_4 + 4H_2O + 5I_2, \text{ in acid solution,}$$

so that the iodine liberated from a neutral solution of KI by a mixture of chlorine and chlorine dioxide represented all the former and one-fifth of the latter.

This work entailed the preparation of pure chlorine dioxide by the method of King & Partington (1926) and also the preparation of solutions containing a known weight of the gas.

Analysis figures for chlorine and chlorine dioxide, expressed in gram molecules of gas per gram-molecule ammonium perchlorate decomposed, as well as percentage weight lost due to decomposition and sublimation, are given in table 2.

Nitrous oxide, chlorine and nitrosyl chloride

Fortunately for the previous scheme, there was no nitrosyl chloride produced below 300° C, while over 380° C the amount of chlorine dioxide was small. This small amount of the dioxide was neglected in the present scheme, which accounts for the fact that the total chlorine figures are slightly higher than theoretical. Unfortunately, there was no stoicheiometric equation for the reaction between nitrosyl chloride and potassium iodide solutions though the possibility was investigated.

The scheme of analysis for these three gases depended upon their different reaction with mercury. Nitrosyl chloride reacts liberating an equivalent amount of nitric oxide, chlorine and its dioxide are completely absorbed giving chloride and chlorate and nitrous oxide has no reaction. The scheme was based upon 'a new method for the analysis of mixtures of chlorine, phosgene and nitrosyl chloride', by Steacie & Smith (1938). The gas pressures were measured using a glass spiral gauge as a null-point instrument in conjunction with a balancing mercury manometer. When nitrosyl chloride was present the nitric oxide was converted to nitrogen tetroxide by adding excess oxygen and measuring the decrease in volume (allowing 10 % dissociation of N_2O_4 at room temperature). Analysis results obtained with this apparatus are given in tables 3 and 4 which also includes some results obtained in the high temperature range of 380 to 450° C.

Perchloric and hydrochloric acids

Both these acids were formed in the decomposition at all temperatures, the total $[H^+]$ varying little in the low-temperature range (220 to 300° C), but being twice as large in the high (above 380° C). At the low temperature there was more perchloric acid than hydrochloric, while at the higher temperatures (380 to 450° C) the reverse was true.

The acids, water and any nitrogen tetroxide formed were found in the first trap after removal of the other gases. The presence of nitrogen tetroxide was shown with Griess reagent (sulphanilic acid and α-naphthylamine in 10 % acetic acid), though its amount was never great enough to warrant determination. The hydrochloric acid was determined as silver chloride and the total hydrogen ion by iodometric titration, giving the perchloric acid as the difference between the two figures. A value for the water produced could be found by difference if required. Some figures for total acidity, etc., are included in tables 2, 3 and 4.

TABLE 3

run no.	temperature (° C)	duration (min)	% decomposed	g mol. O_2	g mol. N_2	N_2O	$Cl_2 + ClO_2$	total H^+
140	225	240	30	0·51	0·126	0·362	0·426	0·155
206	230	170	31·7	—	—	0·35	0·437	0·166
208	230	170	33·8	0·48	0·12	0·32	0·403	0·145
210	230	200	32·5	0·48	0·12	0·313	0·44	0·164
154	235	225	31·9	0·51	0·09	0·34	—	0·12
155	235	280	31·9	0·51	0·09	0·357	0·43	0·166
142	245	250	28·7	0·554	0·061	0·389	0·417	0·147
143	245	200	29·0	0·552	0·061	0·385	0·429	0·135
144	255	180	28·8	0·568	0·063	0·368	0·425	0·131
145	255	200	29·0	0·566	0·063	0·379	0·417	0·143
146	260	175	29·0	0·558	0·062	0·380	0·428	0·149
147	260	190	28·2	0·576	0·064	0·402	0·412	0·156
148	265	140	28·6	0·567	0·063	0·391	0·387	0·146
149	265	120	28·6	0·567	0·063	—	—	0·14
150	270	120	27·8	0·573	0·064	0·44	0·411	0·16
151	270	170	30·6	0·527	0·059	—	0·408	0·121
156	270	90	28·6	0·567	0·063	0·467	0·404	0·141
153	275	110	27·7	0·571	0·063	0·437	0·39	0·135

TABLE 4

run no.	temperature (° C)	initial pressure (cm Hg)	g mol. Cl_2	g mol. N_2O	NOCl	total H^+	$HClO_4$	HCl
P 1	380	1·44	0·293	0·153	0·231	0·302	—	—
P 2	400	5·2	0·269	0·162	0·296	0·299	—	—
P 3	400	5·2	0·228	0·176	0·278	0·245	—	—
P 11	402	4·78	0·252	0·188	0·298	0·279	—	—
P 4	420	4·7	0·24	0·228	0·233	0·255	—	—
P 5	420	3·8	0·251	0·220	0·258	0·258	0·104	0·154
P 12	420	4·0	0·28	0·23	0·205	0·297	—	—
P 8	440	5·3	—	—	—	0·281	—	—
P 10	440	5·91	0·285	0·238	0·164	0·302	0·126	0·176
P 6	450	5·8	0·308	0·22	0·175	0·277	0·112	0·165
P 7	450	4·73	0·29	0·24	0·187	0·273	0·114	0·159

Discussion of analytical results

The most important of the gas analysis results are those relating to oxygen and nitrogen produced in the low-temperature range. The total amount of permanent gas produced per 10 mg of salt decomposed was almost constant at 1·26 ml. at n.t.p. which meant that the pressure-time curves represented the progress of the solid decomposition. The ratio of oxygen to nitrogen was found to be the same over a run at any one temperature, the same whether the gas was allowed to accumulate in the apparatus or pumped off all the time, and, also, nearly unchanged from the value of 9:1 over a large temperature range (240 to 330° C). These results meant that in the accumulatory pressure runs the quantity of nitrogen and oxygen could be calculated simply from the final pressure, provided the volume of the apparatus was known.

3. The type of decomposition curve obtained

The progress of the decomposition was followed by measuring the increase in pressure with time of the permanent gas products. The curves obtained in the temperature range 220 to 280° C were of the sigmoid type usually associated with solid decomposition reactions despite the fact that the decomposition had ceased when only 28 to 30 % of the material had been decomposed. The curves always showed an induction period, during which the reaction was proceeding too slowly for measurement, the length of which depended on temperature and past history

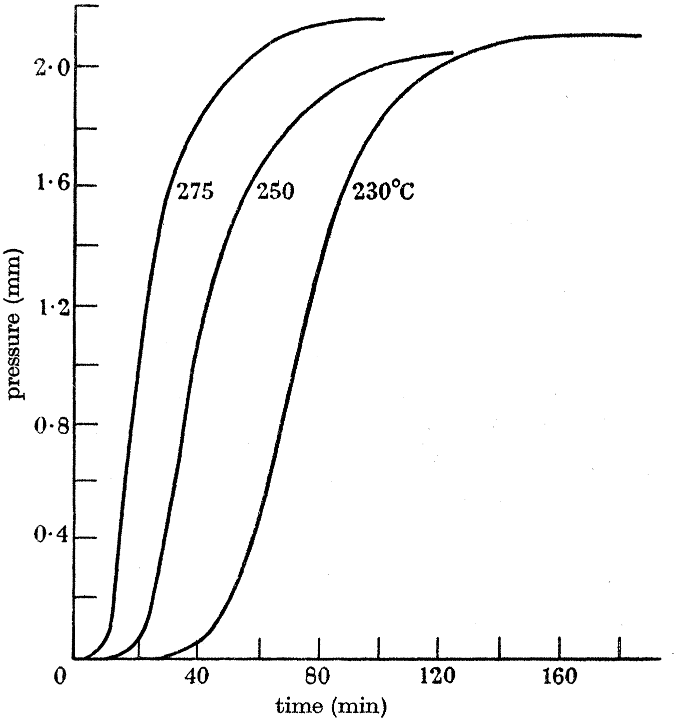

Figure 1. Pressure-time curves.

of the sample. The types of sigmoid curves are indicated in figure 1. After the induction period the rate accelerated rapidly to a maximum value and then fell off again. The deceleratory period was always larger than the acceleratory. The time to the maximum rate also decreased with increasing temperature until for the decompositions measured at 380 to 450° C the curves were nearly 100 % deceleratory.

(i) *Sublimation*

In common with many other ammonium salts, the perchlorate was found to sublime unchanged *in vacuo* at all temperatures at which the decomposition was investigated. It also sublimed at atmospheric pressure if heated in a stream of inert gas, though under static conditions small pressures of permanent gas considerably reduced the amount. The amount of sublimation was not so large in the gas stream as *in vacuo* at the same temperature; e.g. at 260° C there was hardly

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any sublimation in the first case, whereas *in vacuo* with constant pumping it accounted for a loss of 22 % of the original weight after $4\frac{1}{2}$ h heating.

Sublimation was found to be independent of the decomposition process and only dependent on temperature and, as indicated above, to a certain extent on pressure. The effect of temperature on sublimation *in vacuo* (i.e. with pumps running all the time) is shown in table 5. At any one temperature slight variations in the amount of sublimation were found with different runs. The salt *A* (powdered) was heated in 100 mg amounts in the long decomposition tubes previously described in § 1.

TABLE 5

temperature (° C)	226	228	234	235	236	238	241	244	250
% sublimation	4·4	7·2	9·5	7·3	10	6·8	7·5	9·0	17
duration (min)	280	270	270	270	270	270	270	270	240
temperature (° C)	251	253	256	260	276	285	305	327	331
% sublimation	11	16	24	22	37	42	81	85	86
duration (min)	270	270	270	270	270	270	270	90	90

The effect of small initial pressures of oxygen in reducing sublimation is shown in table 6.

TABLE 6

type of run	pressure (mm) initial	pressure (mm) final	duration (min)	% sublimed	% decomposition
continuous pumping	—	—	270	22·5	26
accumulative	10^{-4}	1·2	230	9·6	28·7
accumulative	1·04	2·2	270	4·7	27
accumulative	1·94	3·1	270	2·9	28

Though the amount of sublimation showed a steady increase with increasing temperature (*in vacuo*) the amount of decomposition remained at about 28 % up to 280° C, when it fell off as the salt sublimed at such a fast rate. This, and the fact that sublimation continued after the 28 % decomposition had occurred, was evidence that the decomposition did not take place in the vapour phase.

While the ordinary chemical analysis for ammonium ion, and the X-ray powder spectra, showed the sublimate to be pure ammonium perchlorate of the same crystal form as salt *A*, the results from experiments where the sublimate itself was reheated, and found to decompose with a much reduced induction period, showed that it must contain some trace of a positive catalyst. From spot tests on sublimed material it was shown to contain traces of hydrogen ion, nitrate and nitrite ions. No trace of chloride was ever found in the sublimate.

Experiments carried out at 230° C showed that the sublimate decomposed after an induction period of 10 min which for salt *A* was 25 to 30 min. The subsequent maximum rate was no greater, so that the impurities only affected the induction period. The sublimed material never decomposed to quite the same extent as salt *A*, 24 to 26 % against 28 to 30 %. The catalytic effect of these trace impurities also lowered the temperature at which the salt decomposed in a conveniently measurable time; at 210° C, whereas salt *A* showed no measurable loss in weight after 3 h heating, the decomposition of the sublimate was complete after 5 h.

Recrystallization of sublimate from water (complete slow evaporation of the solvent so that no material was lost but a new crystal lattice was formed) gave crystalline material which, though having a shorter induction period (20 against 30 min), gave a similar pressure-time curve with the same amount of final decomposition as salt *A*.

(ii) *The residue left in the low-temperature decomposition*

Ammonium perchlorate only decomposed to the extent of 28 to 30 % when heated in an inert gas stream or *in vacuo* at temperatures below 290° C. The remaining material, or residue, was no longer of crystalline appearance, but porous. After this amount of decomposition, and provided the residue was not exposed to solvent vapour, further heating caused no loss in weight, except if the pressure were low enough, that due to sublimation.

There were two possibilities for this stoppage of decomposition: either there was chemical poisoning of the decomposing surface by adsorption of one or more of the product gases or it was necessary that a certain lattice configuration be present for the decomposition reaction to be able to compete with the sublimation reaction. Of the two possibilities the first was shown experimentally to be quite incorrect and the second, while being discussed in greater detail later, was incapable of experimental investigation.

Spot tests on the residue after a decomposition did show a trace of nitrogen tetroxide and chlorine to be present, though treatment of salt *A*, or part decomposed salt, with the product gases singly or together, with the exception of water vapour, for varying times and pressure of application, caused no subsequent 'poisoning' effect on the sample when heated.

Recrystallized residue, showing the presence of a trace of nitrite ion, decomposed at 230° C with an induction period of 20 min, which was similar to that obtained with recrystallized sublimate. This treatment, of course, destroyed the porous structure and reformed the crystalline one which allowed further decomposition to occur.

The residue from a low-temperature run was found to decompose completely if heated over 400° C with a small pressure of inert gas to stop sublimation. Salt *A* also decomposed completely under these conditions. This indicates a different mechanism for the decomposition reaction in the two temperature regions.

4. The effect of solvent vapours on the residue

The effect of water vapour, investigated in the same way as that of the other gaseous products, on the partly or completely decomposed material was one of 'rejuvenation' and on reheating further decomposition occurred, depending on the duration of the exposure, and the pressure of the water vapour. It was thus apparent why residues left open to the atmosphere for some time should decompose again on reheating. The effect was attributed to the reformation of a crystalline lattice, free from approximately 30 % voids, which would allow the decomposition reaction to proceed, and the effect of other solvents was investigated.

The extent of 'rejuvenation' was now found to be proportional to the solubility of the perchlorate in the solvent at 25° C, as well as the exposure time. Solubility of the perchlorate in grams per 100 g solvent at 25° C was

water	methyl alcohol	acetone	ethyl alcohol	ethyl acetate
19·95	6·41	2·21	1·87	0·032

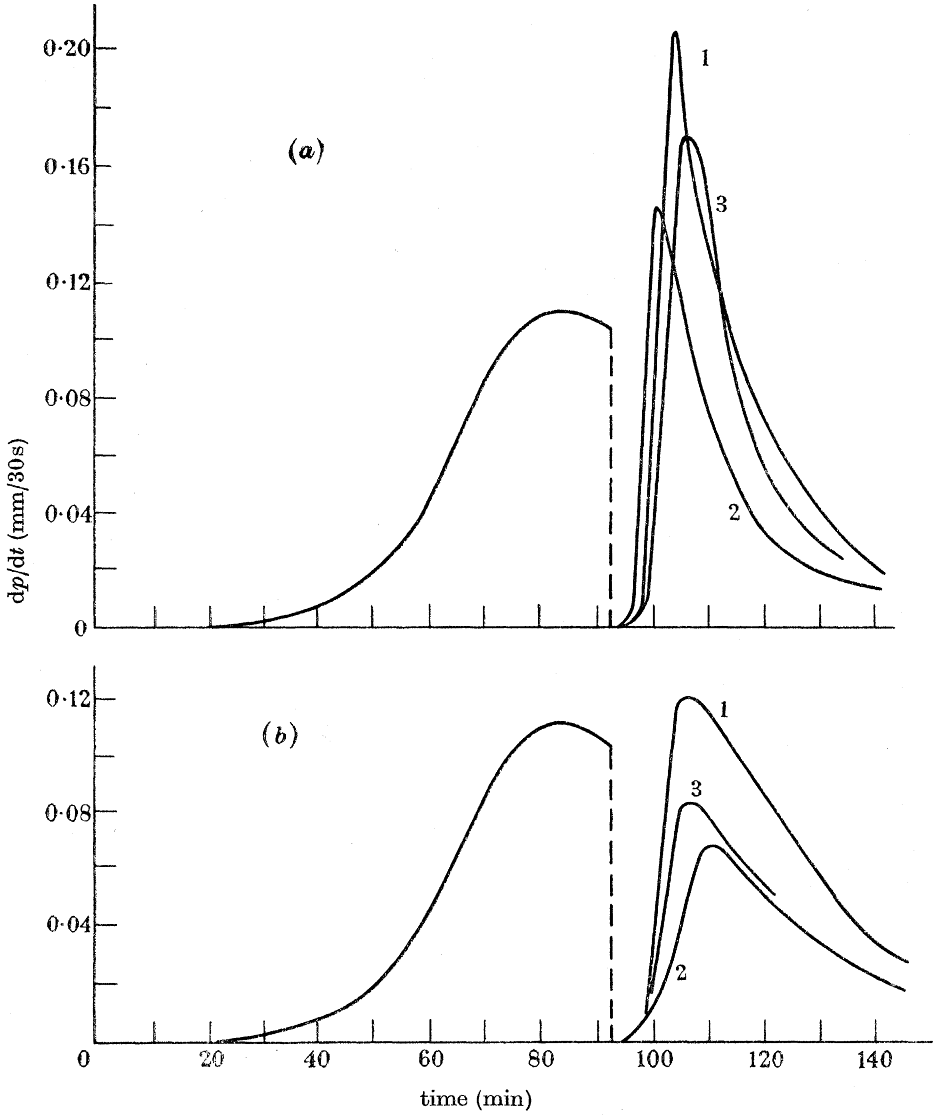

FIGURE 2. Effect of solvent vapours on partly decomposed salt *A* at 230° C. (*a*) 1, H_2O, 50 min at 25 mm; 2, H_2O, 10 min at 25 mm; 3, *Me*OH, 50 min at 81 mm. (*b*) 1, $(Me)_2CO$, 50 min at 68 mm; 2, *Et*OH, 60 min at 40 mm; 3, *EtAc*, 50 min at 66 mm.

The method of determining the extent of the 'rejuvenation' was similar to that used for investigating the effect, if any, of possible poisoning gases. 150 mg of the salt was placed in a short boat (4 cm long × 4 mm internal diam.) with a magnetic pusher attached. It was heated *in vacuo* at 230° C (temperatures above 240° C were not suitable for this investigation because of the complicating factor of the crystal transformation), and the decomposition followed by the rate method described in §1. After 90 min, when the maximum rate has just been passed, the boat was withdrawn from the oven. The decomposition chamber was then turned off from the pumps and on to a trap containing solvent, in equilibrium with its vapour at the required vapour pressure. After 50 min exposure the vapour was recondensed in the trap, the decomposition chamber turned back to the pumps, and heating recommenced. Rate measurements were now taken every $2\frac{1}{2}$ min. The rate rose higher than the previous maximum, with water and methyl alcohol vapours, as high with acetone, and with ethyl alcohol and acetate, whose solvent properties were low, no rise at all was found. Graphs illustrating this solvent effect are shown in figure 2.

These results were interpreted as follows: assuming the sample being heated consisted of particles as pictured in figure 1, part II (Bircumshaw & Newman 1955), then when the sample was withdrawn its cross-section would be similar to figure 1*c*, with decomposition occurring along the interface between residue and undecomposed material. The effect of the solvent vapour was to cause slight recrystallization in the residue so that on reheating decomposition started immediately at the old interface and also in the rejuvenated areas.

Because the new interface between recrystallized and porous material would be relatively large the rate, on reheating, was higher than the previous maximum but only for a short time because of the thinness of the 'rejuvenated' layer. The height of the new maximum rate gave a qualitative picture of the extent of 'rejuvenation' which was in agreement with the solubility of ammonium perchlorate in the various solvents at 25° C.

5. The effect of the crystal transformation on the decomposition curves

At 240° C, ammonium perchlorate undergoes a transformation from orthorhombic to cubic, and this has been found to have a marked effect on the maximum rate of decomposition above and below this temperature. Figure 3 shows how the maximum rate does not increase steadily with rising temperature over the range 224 to 285° C. It rose steadily to a maximum at 238° C, then fell to a minimum at 250° C, and finally rose again. The induction periods and amounts of sublimation did not follow this trend, the former decreasing and the latter increasing steadily with rise in temperature. The possible reasons for this effect are discussed later.

The effect on the rate of decomposition *in vacuo*, of interrupting heating of a sample of perchlorate, both in its acceleratory and deceleratory periods, was investigated above and below the transition temperature. The sample was withdrawn from the furnace, usually for 10 min, by using the short boat already described with magnetic pusher attached. During the interruption it was allowed to cool at room temperature *in vacuo*.

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Below 240° C this procedure was found to have no effect on the subsequent rate, which, after a slight 'warming-up' induction period, returned to its previous value. This lack of a second induction period showed the reaction to be autocatalytic at the interface between the undecomposed and decomposed material.

Above 240° C, in the cubic form, the rate always rose to a much higher value after the cooling period. If the interruption was made in the acceleratory period the maximum rate on reheating was two to three times the normal value, though after reaching this peak value it quickly fell off again. At temperatures not too far removed from the transition, up to 255° C, the rate curve, after an interruption, showed two maxima and a minimum value. This was explained by the proximity to the transition temperature and because the sample did not immediately reach the required temperature after the interruption.

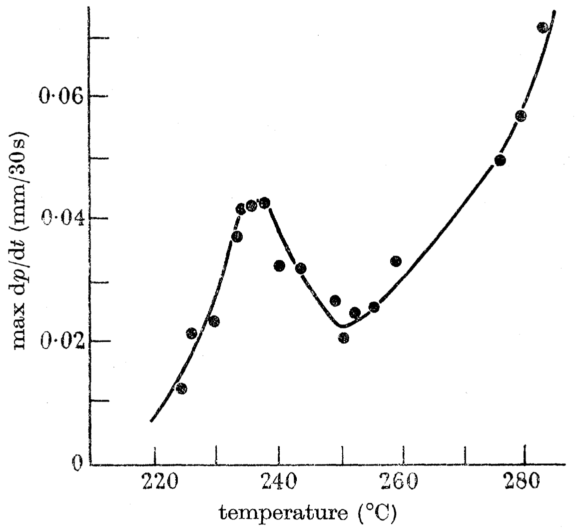

FIGURE 3. Variation of maximum decomposition rate with temperature.

As already stated, between 240 and 255° C, the maximum rate was lower than just before the transition. When a sample was cooled through the transition temperature the change in crystal form must cause the cracking of small crystallites, thus exposing fresh surfaces for decomposition, and which, on reheating, decompose quicker below 240° C. Thus the first peak value which occurred on reheating was due to the immediate decomposition in the orthorhombic form occurring at the old and new interfaces. This peak value passed quickly and the rate was beginning to fall when the temperature of the sample again passed through the transition value and a further set of fresh interfaces appeared. This accounted for the second peak and also for the relative heights of the two peaks.

At 242° C (figure 4*a*), where the normal rate was only slightly slower than at 238° C, the peaks were the same. At 245° C (figure 4*b*), where the maximum rate in the cubic form is at its lowest, the second peak was very small. At 250° C (figure 4*c*) they were again the same. At 255° C (figure 4*d*) that due to the orthorhombic form was small, but it must also be taken into account that the time to pass through the transition temperature, after reheating commenced, diminished with rising temperature. Thus at higher temperatures the two peaks were superimposed giving a very high value. At 260° C the 'orthorhombic contribution'

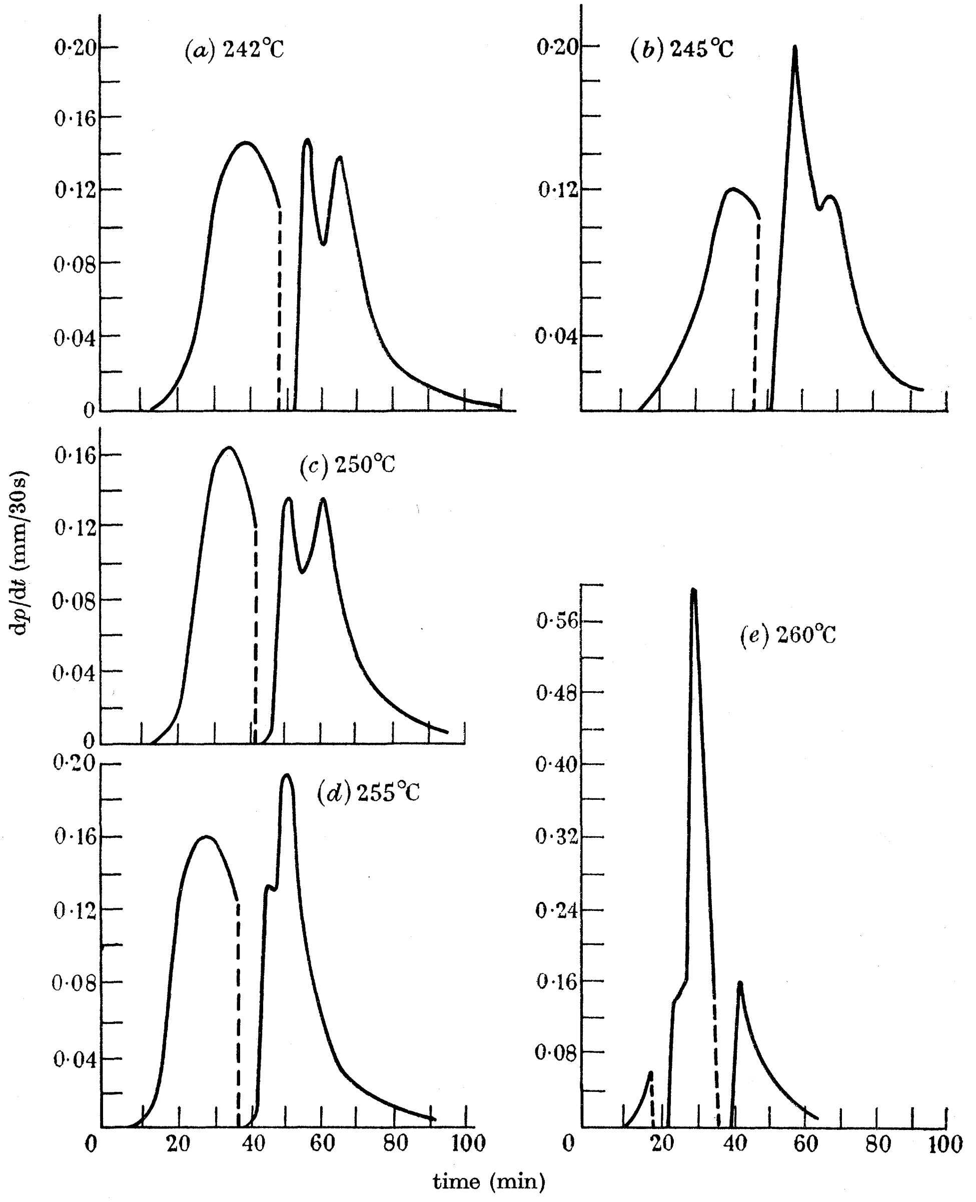

FIGURE 4. Interrupted rate curves.

was only visible as a point of inflexion on the left-hand side of the peak (figure 4*e*).

Though the maximum value for the rate could be much increased by this method, the deceleration was then much faster and the overall amount of decomposition no more than the usual 28 to 30 %.

6. The effect of added impurities on the decomposition

(a) *Ammonium nitrate*

The probability that ammonium nitrate would be the final product in the low-temperature decomposition of all ammonium salts of sufficiently oxygenated acids has been put forward by Dodé. With both ammonium permanganate (Bircumshaw & Tayler 1950) and ammonium chlorate (Fairbrother 1922) this is the case at low temperatures. The decomposition temperature of the perchlorate precludes the formation of the nitrate though it may occur as an intermediate stage. As the presence of trace amounts of nitrate ion in the sublimed material had been established, it was decided to compare the decomposition of the sublimate with that of a sample of perchlorate crystallized out with ammonium nitrate as an impurity. The material was prepared in two ways:

(i) by allowing a mixed solution of the perchlorate and nitrate, in the ratio of 100:1 by weight, to evaporate slowly to dryness, and

(ii) using a similar solution but removing the first crop of crystals and washing free of mother liquor with distilled water.

Spot-test analysis showed no nitrate to be present in the second case, so that the nitrate crystallized out on the surface of the perchlorate. However, at 230° C, both samples showed a reduction of induction period compared with salt *A*; (i) to 15 min, (ii) to 20 min, whereas *A* was 30 min.

(b) *Perchloric acid*

Perchloric acid as a product and the presence of hydrogen ion on the sublimate have been confirmed. It is also quite conceivable that the formation of centres of decomposition on the crystal surface may be associated with the formation of free perchloric acid so that the introduction of free perchloric acid should lower the induction period.

A sample containing a 1 in 50 (by weight) impurity of perchloric acid was prepared by the first of the above methods and the induction period reduced to 15 min at 230° C, though the maximum rate following was not so great as with the sublimate or material containing ammonium nitrate.

(c) *Ammonia*

If the first step in the decomposition was as above, then the presence of any excess of ammonia in the crystal should lengthen the induction period. The perchlorate was recrystallized from 0·880 ammonia in which it is quite soluble. The induction period at 230° C was lengthened to 50 min and the subsequent rate also much lower than with salt *A*, though the usual 28 % decomposition finally occurred.

The effect of a pressure of 3·5 cm anhydrous ammonia, during the first 20 or 40 min heating at 230° C, on the subsequent decomposition rate was investigated. At the end of the exposure time the ammonia was recondensed in a trap at −180° C and the usual rate readings taken. The rate after the exposure was much less than normal though the effect wore off after about 30 min when the sample decomposed as usual (see figure 5*a*). Heating a sample for 2 h in a pressure of 3·5 cm ammonia at 230° C reduced the decomposition from about 30 % to 3 %.

(*d*) *Ammonium chloride*

It had been reported by Saunders (1922) that the presence of small amounts of sodium or ammonium chlorides greatly accelerated the thermal decomposition of ammonium nitrate. However, a 1 in 250 addition of ammonium chloride to the perchlorate, prepared by the first method, lengthened the induction period by 10 min at 230° C.

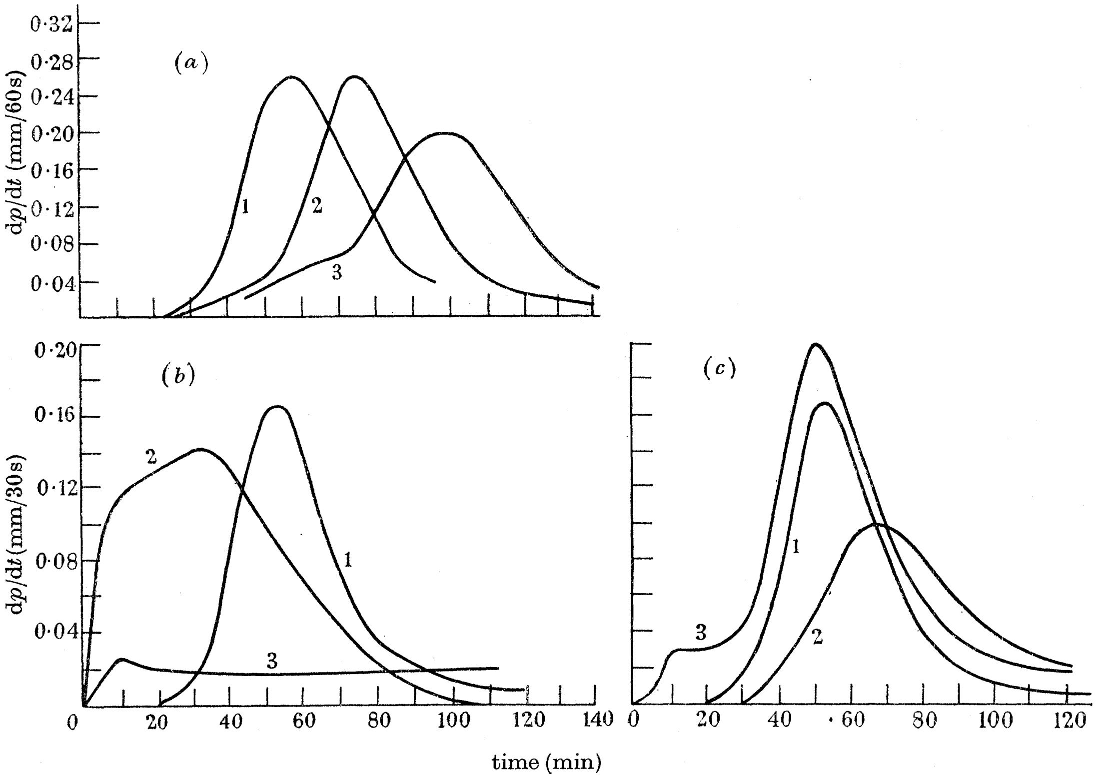

FIGURE 5. (*a*) Effect of ammonia during induction period: 1, no ammonia; 2, 3·5 cm NH_3 for 20 min; 3, 3·5 cm NH_3 for 40 min. (*b*) Effect of addition of 10 % MnO_2: 1, 230° C, pure NH_4ClO_4, 29 % decomp.; 2, 230° C, +10 % MnO_2, 100 % decomp.; 3, 200° C, +10 % MnO_2, 25 % decomp. (*c*) Effect of addition of 10 % Fe_2O_3 and CaO: 1, pure NH_4ClO_4, 29 % decomp; 2, +10 % Fe_2O_3, 43 % decomp.; 3, +10 % CaO, 25 %. Temperature 230° C.

7. THE EFFECT OF ADDED METALLIC OXIDES

Little work was done on the catalyzed decomposition of ammonium perchlorate. Additions of 10 % by weight of Al_2O_3, CaO, Fe_2O_3 and MnO_2 were made to salt *A* and their effect determined by following the rate at 230° C and determining the amount of decomposition after 2 h.

Manganese dioxide had by far the most pronounced catalytic effect; 100 % decomposition, with no sublimation, occurred *in vacuo*, and the rate curve showed a similarity to the high-temperature decomposition at 400° C, though a large amount of chlorine dioxide, and not nitrosyl chloride, was produced. Even at 200° C, where the pure salt *A* would not have decomposed at all *in vacuo*, in the presence of manganese dioxide, 25 % decomposition occurred in 2 h. There was no sublimation, or stoppage of decomposition, and again a large amount of chlorine dioxide was formed (figure 5*b*). Ferric oxide also had a catalytic effect. In 2 h at 230° C, 44 % had decomposed and in 4 h 60 %. The rate curve was similar to that for salt *A* at 230° C superimposed on that with manganese dioxide added at 200° C. Thus there was no induction period, the rate reaching a steady value almost immediately and then after 25 min increasing again and giving the usual type of curve. The deceleratory portion did not, however, return to zero rate but steadied off a little lower than the earlier steady value; this latter represents the decomposition due to the catalyst only, while the peak value was the usual curve at 230° C.

Aluminium oxide had no effect at all, decomposition ceasing in the usual way at 28·5 %. Calcium oxide, which would encourage the formation of ammonia, slowed the reaction down to 25 % decomposition in the 2 h (figure 5*c*).

References

Bircumshaw, L. L. & Newman, B. H. 1955 *Proc. Roy. Soc.* A, **227**, (in the press) (part II).
Bircumshaw, L. L. & Tayler, M. 1950 *J. Chem. Soc.* p. 3674.
Dodé, M. 1935 *C.R. Acad. Sci., Paris*, **200**, 63.
Dodé, M. 1938 *Bull. Soc. chim Fr.* (5) **5**, 170.
Fairbrother, F. 1922 *J. Amer. Chem. Soc.* **44**, 2419.
King, E. F. & Partington, J. R. 1926 *J. Chem. Soc.* p. 925.
Martin, G. G. & Woolaver, L. B. 1945 *Industr. Engng Chem.* (Anal. ed), p. 474.
Naoum, P. & Aufschlager, R. 1924 *Z. ges. Schiess- u. Sprengstoffw.* **19**, 121.
Saunders, H. L. 1922 *J. Chem. Soc.* **121**, 698.
Smeets, C. 1934 *Natuurwet. Tijdschr.* **16**, 262.
Steacie, E. W. R. & Smith, W. M. 1938 *Canad. J. Res.* B, **16**, 1.